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siddharth@siddharthreddy:~/OSLAB$ nano m23.c
siddharth@siddharthreddy:~/OSLAB$ gcc m23.c -o m23
siddharth@siddharthreddy:~/OSLAB$ ./m23
=====
COMMAND HISTORY BUFFER
=====

Commands:
  add <command>    - Store command
  history          - Display history
  get <number>     - Retrieve command
  clear           - Clear history
  exit            - Exit

history-shell> hi
Invalid command.

history-shell> add
Invalid command.

history-shell> clear
History cleared.

history-shell> |
```

```
siddharth@siddharthreddy:~$ cd OSLAB
siddharth@siddharthreddy:~/OSLAB$ nano m11.sh
siddharth@siddharthreddy:~/OSLAB$ chmod +x m11.sh
siddharth@siddharthreddy:~/OSLAB$ ./m11.sh
=====
      SINGLE QUOTES AND DOUBLE QUOTES
=====

----- SINGLE QUOTES -----
Hello World
Hello $NAME
The value of $NAME is not expanded.
Stored single-quoted string:
This is a $NAME literal string.
Literal content:
Hello      World      Linux

----- DOUBLE QUOTES -----
Hello World
Hello Shanmukha
The value of NAME is: Shanmukha
Stored double-quoted string:
Hello Shanmukha
String with spaces:
Hello      Linux      World

----- VARIABLE EXPANSION -----
Single quotes: $NAME
Double quotes: Shanmukha

----- NESTED TOKENS -----
Outer quotes with variable: Linux Shell
Command result: Nested Command

----- QUOTED COMMANDS -----
Command stored as:
echo Hello Linux

Executing command:
Hello Linux

----- EDGE CASES -----
Length of single-quoted empty string: 0
Length of double-quoted empty string: 0
Special characters: $ * ? ! # @
Special characters: $ * ? ! # @
```

```
All quote tests completed successfully.
siddharth@siddharthreddy:~/OSLAB$ nano m12.sh
siddharth@siddharthreddy:~/OSLAB$ bash m12.sh
=====
          ESCAPE SEQUENCE DEMONSTRATION
=====

----- ESCAPED SPACES -----
Hello Linux
Hello Linux World

----- ESCAPING SPECIAL SYMBOLS -----
Dollar symbol: $
Asterisk: \*
Question mark: \?
Exclamation mark: \!
Hash symbol: \#
Backslash: \

----- ESCAPED CHARACTERS -----
Line1
Line2
Column1 Column2
FirsSecond
World

----- PRESERVING CHARACTERS -----
Hello $USER \* \? \#

----- ESCAPED FILE NAME -----
Created file:
-rw-r--r-- 1 siddharth siddharthreddy 0 Oct  1 18:12 'My Test File.txt'

----- COMPLEX INPUT -----
Hello\ World\ $USER\ \*\ \?

----- PARSER OUTPUT -----
Normal:
Hello World
Escaped:
Hello World
Quoted:
Hello World

Escape sequence testing completed.
```